HeatGuard: A Rule-Based Expert System for Heatwave Early Warning and Stakeholder Advisory

LAB CA Component 2 / Experiment 10: Design and Implementation of a Simple Rule-Based Expert System for a Real-World Advisory Problem

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D1. Problem Definition

1.1 Domain and motivation

Heatwaves are among the deadliest weather hazards in India. With climate change they are becoming more frequent, more intense and longer. The SVU use case **KJS-CES-01** sets out a five-phase heatwave intelligence platform: data acquisition, AI forecasting, IoT weather stations, forecast validation, and **Phase V: Advisory, Early Warning and Decision Support**. Phase V must "translate technical forecasts into stakeholder-specific advisories and early warnings for citizens, farmers, health agencies, and local authorities".

HeatGuard is a rule-based expert system for Phase V. It takes a temperature observation or forecast for a location, which in the full platform would come from IMD data, an AI forecast or an IoT Automated Weather Station. From that input it:

1. classifies the day using the **official IMD heat-wave criteria** (none / heat wave / severe heat wave);
2. assigns an **IMD colour-coded alert** (Green / Yellow / Orange / Red), taking persistence into account;
3. **escalates** the alert when nights are warm or humid heat is dangerous;
4. **triages heat-related illness symptoms** (heat stroke vs. heat exhaustion vs. heat cramps);

5. issues **advisories tailored to the stakeholder** (citizen, vulnerable person, outdoor worker, farmer, health department, municipal authority);
6. shows a full **explanation trace**: which rules fired, why, and on which guideline each rule is based.

1.2 Why a rule-based expert system?

Heat-wave declaration in India is *defined* by explicit, published thresholds. A heat wave is not a fuzzy pattern to be learned; it is a regulatory definition. Advisories (NDMA Do's & Don'ts, MoHFW first aid) are also explicit IF-THEN guidance. A rule-based system is therefore:

- **faithful**: it reproduces the official criteria exactly;
- **transparent and auditable**: the use case marks *Responsible AI* and *Human-in-the-loop* as **High** relevance, and every conclusion here can be traced to a rule and its source;
- **complementary to ML**: Phase II forecasts *what the temperature will be*; this system decides *what that temperature means and what people should do*. It can also check the facts in LLM-generated advisories (use-case governance item "ensuring factual consistency").

1.3 Scope

In scope	Out of scope
IMD heat-wave & severe heat-wave classification for plains, coastal and hilly stations	Forecasting temperature (Phase II ML models)
IMD colour-coded alert from persistence (days)	Multi-station spatial aggregation (IMD needs ≥2 stations in a sub-division)
Warm-night and humid-heat (heat index) escalation	Real-time AWS ingestion (inputs are typed in or come from JSON)
First-level triage of heat-related illness symptoms	Medical diagnosis; the system only advises calling 108/112
Advisories for 4 stakeholder roles and 5 vulnerable groups	Multilingual / LLM text generation

1.4 Target users

- **Citizens and households**, including the elderly, infants, pregnant women, the chronically ill and outdoor workers.
- **Farmers** (crops and/or livestock).
- **Public health departments and hospitals** (preparedness and response).
- **Municipal corporations / District & State Disaster Management Authorities** (Heat Action Plan triggers).
- **Duty meteorologists**, who review Red alerts (human in the loop).

1.5 Sources of expert knowledge

The rules are taken from published guidelines, not invented:

- **S1** - IMD - Heat wave criteria (FAQ on Heat Wave, mausam.imd.gov.in): threshold 40 C plains / 37 C coastal / 30 C hilly; departure from normal 4.5-6.4 C = heat wave, >6.4 C = severe; actual Tmax >=45 C = heat wave, >=47 C = severe; declared when met on 2 consecutive days.

- **S2** - IMD - Colour-coded impact-based heat warnings (Green / Yellow / Orange / Red) based on intensity and persistence of heat wave days.
- **S3** - IMD - Warm night criteria: Tmax >=40 C and Tmin departure 4.5-6.4 C = warm night, >6.4 C = very warm night.
- **S4** - NDMA - Guidelines for Preparation of Action Plan: Prevention and Management of Heat Wave (2016, rev. 2019): Do's & Don'ts, stakeholder roles, cooling centres, work-hour rescheduling.
- **S5** - MoHFW / NCDC - National Programme on Climate Change & Human Health (NPCCHH): National Action Plan on Heat Related Illnesses (2021) - HRI symptoms, first aid, hospital preparedness.
- **S6** - US NWS / NOAA - Heat Index (Rothfusz regression, 1990) and risk bands: Caution 27-32 C, Extreme caution 32-39 C, Danger 39-51 C, Extreme danger >=51 C.
- **S7** - IMD / ICAR - Gramin Krishi Mausam Sewa agromet advisories for heat wave: irrigation timing, mulching, spraying, livestock care.
- **S8** - Ahmedabad Heat Action Plan (AMC, 2013 onward) - model city plan for alert-triggered actions by municipal and health departments.
- **S9** - SVU use case KJS-CES-01 - AI governance: human-in-the-loop review of critical warnings before dissemination.

D2. Knowledge Base Design

2.1 Facts (working-memory attributes)

Fact	Type / values	Origin
`role`	citizen, farmer, health, authority	user
`region`	plains, coastal, hilly	user
`tmax`, `tmax_normal`	°C	user (`tmax_normal` optional)
`heat_days`	1-60 consecutive days	user (optional)
`rh`	0-100 %	user (optional)
`tmin`, `tmin_normal`	°C	user (optional, asked only if Tmax ≥ 40)
`vulnerable`	elderly, infant_child, pregnant, chronic_illness, outdoor_worker	user (citizen)
`farm`	crops, livestock, both	user (farmer)
`symptoms`	12 symptom codes	user (citizen / health)
`departure`, `tmin_departure`, `heat_index`	°C	derived (D1-D3)
`threshold_met`, `hw_category`, `night`, `hi_level`, `hri`	symbolic	inferred
`alert`, `status`, `escalated`, `needs_review`, `recommendation`	symbolic	inferred (conclusions)

2.2 Rule layers and salience

The knowledge base holds **59 production rules** in 9 layers. Salience (priority) follows the order in which a duty officer would reason: life safety first, then data quality, then classification, then advice.

Layer	Rules	Salience	Purpose
T - Triage	T1-T7	107-111	Heat-illness triage; runs even if the weather data are invalid
V - Validation	V1-V5, D0	98-100	Reject impossible inputs (HALT)
D - Derivation	D1-D3	90	Departure from normal, Tmin departure, heat index
C/W/H - Classification	C1-C8, W1-W2, H1-H4	70-80	IMD threshold & category, warm night, heat-stress band
A - Alert	A1-A8	55-62	IMD colour code from category + persistence
E - Escalation	E1-E3	49-50	Warm night / humid heat raises the alert one level
G - Governance	G1-G3	40	Human review of Red alerts; notes on missing data
P/F/M/X - Advisory	14 rules	20	Citizen, farmer, health-department, authority advice
K - Summary	K1	0	Final recommendation

2.3 Complete rule table (IF-THEN)

ID	Rule	IF	THEN	Sal.	Src.
T1	Heat stroke from neurological signs	symptoms contains_any [‘confusion’, ‘unconscious’, ‘seizure’]	hri = ‘heat_stroke’	111	S5
T2	Heat stroke from hot dry skin	symptoms contains ‘high_body_temp’ AND symptoms contains ‘no_sweating’ AND hri is unknown	hri = ‘heat_stroke’	110	S5
T3	Heat exhaustion	hri is unknown AND symptoms contains_any [‘heavy_sweating’, ‘dizziness’, ‘nausea’, ‘weakness’, ‘headache’, ‘fainting’]	hri = ‘heat_exhaustion’	109	S5
T4	Heat cramps	hri is unknown AND symptoms contains	hri = ‘heat_cramps’	108	S5

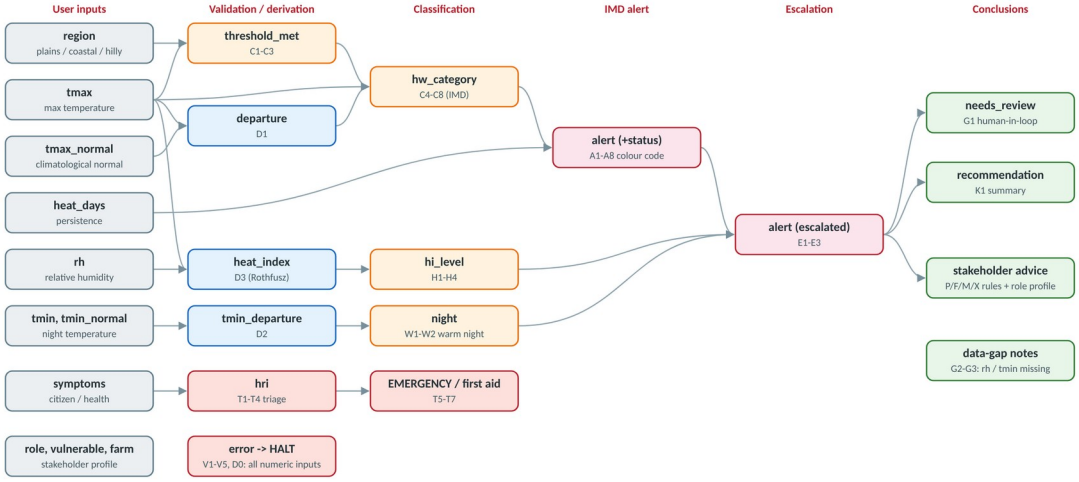
ID	Rule	IF	THEN	Sal.	Src.
		'muscle_cramps'			
T5	Heat stroke emergency advice	hri == 'heat_stroke'	emergency = True; advise EMERGENCY	107	S5
T6	Heat exhaustion first aid	hri == 'heat_exhaustion'	advise FIRST AID	107	S5
T7	Heat cramps first aid	hri == 'heat_cramps'	advise FIRST AID	107	S5
V1	Implausible maximum temperature	tmax outside [-10, 55]	error = 'Tmax outside plausible range -10..55 C'; HALT	100	S1
V2	Implausible normal temperature	tmax_normal outside [5, 50]	error = 'Normal Tmax outside plausible range 5..50 C'; HALT	100	S1
V3	Invalid humidity	rh outside [0, 100]	error = 'Relative humidity must be 0..100 %'; HALT	100	S6
V4	Invalid persistence	heat_days outside [1, 60]	error = 'Consecutive hot days must be between 1 and 60'; HALT	100	S2
D0	Check Tmin against Tmax	tmin is known AND tmax is known	tmin_gt_tmax =	99	S3
V5	Tmin greater than Tmax	tmin_gt_tmax == True	error = 'Minimum temperature is higher than maximum temperature'; HALT	98	S3
D1	Compute Tmax departure from normal	tmax is known AND tmax_normal is known	departure =	90	S1
D2	Compute Tmin departure from normal	tmin is known AND tmin_normal is known	tmin_departure =	90	S3
D3	Compute heat index	tmax is known AND rh is known	heat_index =	90	S6
C1	Plains threshold reached	region == 'plains' AND tmax >= 40	threshold_met = True	80	S1
C2	Coastal threshold reached	region == 'coastal' AND tmax >= 35	threshold_met = True	80	S1
C3	Hilly threshold reached	region == 'hilly' AND tmax >= 30	threshold_met = True	80	S1
C4	Severe heat wave by absolute temperature	threshold_met == True AND tmax >= 47 AND hw_category is unknown	hw_category = 'severe_heatwave'	78	S1
C5	Severe heat wave by departure	threshold_met == True AND departure > 6.4 AND hw_category is unknown	hw_category = 'severe_heatwave'	78	S1
C6	Heat wave by absolute temperature	threshold_met == True AND tmax >= 45 AND hw_category is	hw_category = 'heatwave'	76	S1

ID	Rule	IF	THEN	Sal.	Src.
		unknown			
C7	Heat wave by departure	threshold_met == True AND 4.5 <= departure <= 6.4 AND hw_category is unknown	hw_category = 'heatwave'	76	S1
C8	No heat wave (default)	tmax is known AND hw_category is unknown	hw_category = 'none'	70	S1
W1	Warm night	tmax >= 40 AND 4.5 <= tmin_departure <= 6.4	night = 'warm_night'	80	S3
W2	Very warm night	tmax >= 40 AND tmin_departure > 6.4	night = 'very_warm_night'	80	S3
H1	Heat index: caution	27 <= heat_index < 32	hi_level = 'caution'	80	S6
H2	Heat index: extreme caution	32 <= heat_index < 39	hi_level = 'extreme_caution'	80	S6
H3	Heat index: danger	39 <= heat_index < 51	hi_level = 'danger'	80	S6
H4	Heat index: extreme danger	heat_index >= 51	hi_level = 'extreme_danger'	80	S6
A1	Green - no heat wave	hw_category == 'none' AND alert is unknown	alert = 'GREEN'	60	S2
A2	Heat wave on day 1 = watch	hw_category in ['heatwave', 'severe_heatwave'] AND heat_days == 1 AND alert is unknown	alert = 'YELLOW'; status = 'watch'	62	S1
A3	Heat wave persisting 2-3 days	hw_category == 'heatwave' AND 2 <= heat_days <= 3 AND alert is unknown	alert = 'YELLOW'; status = 'declared'	60	S2
A4	Heat wave persisting 4-6 days	hw_category == 'heatwave' AND 4 <= heat_days <= 6 AND alert is unknown	alert = 'ORANGE'; status = 'declared'	60	S2
A5	Heat wave persisting > 6 days	hw_category == 'heatwave' AND heat_days > 6 AND alert is unknown	alert = 'RED'; status = 'declared'	60	S2
A6	Severe heat wave for 2 days	hw_category == 'severe_heatwave' AND heat_days == 2 AND alert is unknown	alert = 'ORANGE'; status = 'declared'	60	S2
A7	Severe heat wave for > 2 days	hw_category == 'severe_heatwave' AND heat_days > 2 AND alert is unknown	alert = 'RED'; status = 'declared'	60	S2
A8	Alert when duration not given	hw_category in ['heatwave',	alert = 'YELLOW';	55	S2

ID	Rule	IF	THEN	Sal.	Src.
		'severe_heatwave'] AND heat_days is unknown AND alert is unknown	status = 'watch'		
E1	Warm night escalates Yellow to Orange	night in ['warm_night', 'very_warm_night'] AND alert == 'YELLOW'	alert = 'ORANGE'; escalated = 'warm night'	50	S3
E2	Humid heat on a non-heatwave day	alert == 'GREEN' AND hi_level in ['danger', 'extreme_danger']	alert = 'YELLOW'; escalated = 'humid heat stress'	50	S6
E3	Extreme heat index escalates Yellow	alert == 'YELLOW' AND hi_level == 'extreme_danger' AND rh >= 40 AND escalated is unknown	alert = 'ORANGE'; escalated = 'extreme humid heat'	49	S6
G1	Red alert needs human validation	alert == 'RED'	needs_review = True; advise GOVERNANCE	40	S9
G2	Humidity missing	rh is unknown AND tmax is known	advise NOTE	40	S6
G3	Night temperature missing	tmin is unknown AND tmax >= 40	advise NOTE	40	S3
P1	Citizen - normal day	role == 'citizen' AND alert == 'GREEN'	advise CITIZEN	20	S4
P2	Citizen - basic heat precautions	role == 'citizen' AND alert in ['YELLOW', 'ORANGE', 'RED']	advise CITIZEN	20	S4
P3	Citizen - severe alert precautions	role == 'citizen' AND alert in ['ORANGE', 'RED']	advise CITIZEN	20	S4
P4	Citizen - vulnerable person at home	role == 'citizen' AND alert in ['YELLOW', 'ORANGE', 'RED'] AND vulnerable contains_any ['elderly', 'infant_child', 'pregnant', 'chronic_illness']	advise VULNERABLE	20	S5
P5	Citizen - outdoor worker	role == 'citizen' AND alert in ['YELLOW', 'ORANGE', 'RED'] AND vulnerable contains 'outdoor_worker'	advise WORKER	20	S4
F1	Farmer - normal day	role == 'farmer' AND alert == 'GREEN'	advise FARMER	20	S7
F2	Farmer - crop protection	role == 'farmer' AND alert in ['YELLOW', 'ORANGE', 'RED'] AND farm in ['crops', 'both']	advise FARMER	20	S7
F3	Farmer - livestock protection	role == 'farmer' AND alert in ['YELLOW',	advise FARMER	20	S7

ID	Rule	IF	THEN	Sal.	Src.
		'ORANGE', 'RED'] AND farm in ['livestock', 'both']			
M1	Health dept - alert preparedness	role == 'health' AND alert == 'YELLOW'	advise HEALTH	20	S5
M2	Health dept - severe alert response	role == 'health' AND alert in ['ORANGE', 'RED']	advise HEALTH	20	S5
M3	Health dept - normal day	role == 'health' AND alert == 'GREEN'	advise HEALTH	20	S5
X1	Authority - heat alert actions	role == 'authority' AND alert == 'YELLOW'	advise AUTHORITY	20	S8
X2	Authority - severe alert actions	role == 'authority' AND alert in ['ORANGE', 'RED']	advise AUTHORITY	20	S4
X3	Authority - normal day	role == 'authority' AND alert == 'GREEN'	advise AUTHORITY	20	S8
K1	Compose final recommendation	alert is known	recommendation =	0	S2

2.4 Rule-dependency diagram



Rule-dependency diagram: how facts flow from user inputs through derivation and classification to the alert and the stakeholder advice.

2.5 Design decisions for consistency and non-redundancy

- **Mutually exclusive categories.** C4-C7 all require `hw_category is unknown`. With severe rules at higher salience (78) than heat-wave rules (76), the first category concluded is the final one, so no two category rules contradict each other.
- **Single default.** C8 (*no heat wave*) has the lowest classification salience (70). It fires only when no criterion matched. This is the expert-system form of "otherwise".

- **Escalation by one level only.** E3 requires `escalated is unknown`, so warm night plus humid heat cannot push Yellow to Red. The use case asks the system to communicate uncertainty, not to over-warn.
- **Edge cases covered.** Exact thresholds (40.0 °C, +4.5 °C) and just-below values (39.9 °C) are tested. So are missing optional inputs, impossible values, physically inconsistent values (Tmin > Tmax), and dangerous humid heat below the IMD thresholds (Mumbai).

2.6 Automated consistency verification

`python -m heatwave_es.kb_check` analyses the knowledge base statically and runs it against every test scenario:

Check	Result
Every fact tested in an IF part is a user input or is asserted by some rule (no dead rules)	0 problems
Every rule cites a known source and has a justification	0 problems
Redundancy: two rules with the same IF *and* the same THEN	0 found
Contradiction: two rules with the same IF, different THEN, same salience	0 found
Rule coverage: rules that fire in at least one test scenario	**59/59 (100 %)**

The checker was itself tested by injecting a duplicate rule, a rule on a non-existent fact, and a contradicting rule. All three defects were reported.

D3. Implementation

3.1 Architecture

```
heatwave_es/
  engine.py          generic forward-chaining engine (working memory,
agenda,
                     conflict resolution, trace, HOW explanation)
  knowledge_base.py  the rules + sources + heat-index computation
(domain only)
  explain.py         recommendation / advisory / trace / HOW
formatting
  cli.py             interactive question-answer interface (WHY, skip,
HOW)
  scenarios.py       test scenarios (D4)
server.py            JSON API + host for the React UI (stdlib only)
frontend/            HeatGuard Atlas React UI (Vite, Tailwind, React
Bits)
app.py               Streamlit web interface (optional)
prolog/heatguard.pl  SWI-Prolog edition: backward chaining meta-
interpreter
tests/               pytest suite
```

The **knowledge base is kept separate from the inference engine**. The rules are data (`Rule` objects with conditions, actions, salience and source), so a domain expert could add a rule without touching the engine.

3.2 Choice of inference strategy: forward chaining (justified)

The Python engine uses **forward chaining** (data-driven), for these reasons:

1. **The data arrive first.** In the use-case pipeline, observations and forecasts (IMD grids, AWS readings, ML forecasts) are pushed to the advisory module. The natural question is "given these readings, what follows?", which is a forward-chaining question.
2. **There are many conclusions, not one goal.** From one observation the system must derive the category, the alert, escalations, governance flags *and* advice for several audiences. Backward chaining would need a separate query for each goal.
3. **Monitoring suits it.** In deployment the same rules re-run every time an AWS reading arrives. Forward chaining is the standard architecture for monitoring and alerting systems (as in CLIPS and OPS5).
4. **Priority matters.** Salience-based conflict resolution lets *life-safety triage* fire before everything else and *validation* halt bad data before any classification.

For comparison, the **Prolog edition uses backward chaining**. It starts from the goal `final_alert(L)` and asks the user *only* for facts that the current line of proof needs. For example, it never asks for the normal temperature when Tmax is below the regional threshold. This is ideal for an interactive consultation, and the viva discussion compares the two.

3.3 The recognise-act cycle and conflict resolution

```
repeat
    MATCH   : agenda = all rules whose conditions hold and that have
not fired
    if agenda empty: stop
    RESOLVE : pick 1 rule by (1) refraction, (2) highest salience,
              (3) highest specificity (most conditions), (4) rule
order
    ACT     : assert/modify facts, emit advice, record trace; HALT if
rule says so
```

- **Refraction**: each rule fires at most once, which guarantees termination.
- **Salience**: domain priority, e.g. T1 (heat stroke, 111) before V1 (validation, 100).
- **Specificity**: a more specific rule beats a general one at equal salience.
- **Robustness**: conditions on missing facts are simply not satisfied, so the engine never crashes on missing input. Implausible values trigger validation rules that HALT with a clear message. A type error inside an action (e.g. text where a number is expected) is caught and reported as an input error.

3.4 Explanation facility

- **Trace**: for every fired rule, the system shows its id, name, salience and source, each condition with the *actual* value matched, the facts asserted, a one-line justification ("WHY"), and the other rules that were in the conflict set at that moment.

- **HOW**: a recursive proof tree for any concluded fact (e.g. `how alert`). It correctly shows an escalated alert as derived from the earlier alert value.
- **WHY (during questioning)**: typing `why` at any question explains why it is asked and lists the rules that use the answer.
- **Sources**: every report ends with the guidelines used for that run.

3.5 User interface

- **CLI** (`python -m heatwave_es`): asks only relevant questions (role-dependent, and Tmin only when Tmax ≥ 40 °C). It accepts prefixes (`p` → plains), validates ranges, lets optional questions be skipped, re-asks politely on bad input, and supports `why`, `how <fact>`, `new` and `quit`.
- **Batch mode** (`python -m heatwave_es --json case.json`) and **rule listing** (`--rules`).
- **HeatGuard Atlas web UI** (`python server.py`, then open http://localhost:8000): a React + Tailwind single-page app built with React Bits components. It has an Aurora hero, a four-step Stepper consultation, the live verdict with a decrypting alert level, SpotlightCards, AnimatedList advisories and rule-firing trace, an interactive HOW proof tree, and a searchable knowledge base that highlights the rules that fired. The Python standard-library server (`server.py`) exposes the engine as a JSON API (`/api/consult`, `/api/kb`, `/api/scenarios`), so the UI and the CLI share exactly the same inference engine.
- **Streamlit UI** (`streamlit run app.py`): a lighter alternative interface.
- **Prolog** (`swipl prolog/heatguard.pl`, then `?- start.`): goal-driven consultation with `why` and a proof tree.

3.6 Sample run (TC09, warm-night escalation)

```
==========================================================================
==
RECOMMENDATION
==========================================================================
==
ORANGE ALERT (Severe heat alert - be prepared) - IMD category:
heatwave
  - Tmax departure from normal: +5.8 C
  - Heat index (feels like): 51.7 C - extreme danger
  - Night: warm night
  - Alert escalated because of: warm night

ADVISORY
--------------------------------------------------------------------------
--
[CITIZEN]
  * Avoid going out in the sun, especially 12:00-15:00.  (P2)
  * Drink water often even if not thirsty; use ORS, lassi, buttermilk,
lemon water, aam panna.  (P2)
  * Wear light, loose, light-coloured cotton clothes; cover head with
cap/umbrella/cloth.  (P2)
  * Avoid strenuous activity outdoors; stay in the coolest room, keep
curtains/shades closed by day, open windows at night.  (P3)
```

```
  * Never leave children or pets in parked vehicles.  (P3)
  * Avoid alcohol, tea, coffee, carbonated drinks and heavy high-
protein meals.  (P3)
[VULNERABLE]
  * Check on elderly, infants, pregnant women and people with
heart/kidney disease or diabetes at least twice a day.  (P4)
  * Keep medicines below 25 C; consult doctor about diuretics / fluid
limits before increasing water intake.  (P4)
[WORKER]
  * Shift heavy outdoor work to early morning / evening; rest in shade
frequently and drink a cup of cool water every 15-20 min.  (P5)
  * Newly returned or new workers: build up exposure gradually over a
week (acclimatisation).  (P5)

EXPLANATION TRACE (15 rules fired)
------------------------------------------------------------------------
--
 1. D0  Check Tmin against Tmax   [salience 99, source S3]
 2. D1  Compute Tmax departure from normal   [salience 90, source S1]
 3. D2  Compute Tmin departure from normal   [salience 90, source S3]
 4. D3  Compute heat index   [salience 90, source S6]
 5. C1  Plains threshold reached   [salience 80, source S1]
 6. W1  Warm night   [salience 80, source S3]
 7. H4  Heat index: extreme danger   [salience 80, source S6]
 8. C6  Heat wave by absolute temperature   [salience 76, source S1]
 9. A3  Heat wave persisting 2-3 days   [salience 60, source S2]
10. E1  Warm night escalates Yellow to Orange   [salience 50, source
S3]
11. P4  Citizen - vulnerable person at home   [salience 20, source S5]
12. P5  Citizen - outdoor worker   [salience 20, source S4]
13. P2  Citizen - basic heat precautions   [salience 20, source S4]
14. P3  Citizen - severe alert precautions   [salience 20, source S4]
15. K1  Compose final recommendation   [salience 0, source S2]

HOW WAS THE ALERT LEVEL CONCLUDED?
------------------------------------------------------------------------
--
alert = ORANGE  <- E1 'Warm night escalates Yellow to Orange'
  night = warm_night  <- W1 'Warm night'
    tmax = 45.6  (given by user)
    tmin_departure = 5.0  <- D2 'Compute Tmin departure from normal'
      tmin = 31.0  (given by user)
      tmin_normal = 26.0  (given by user)
  alert = YELLOW  <- A3 'Heat wave persisting 2-3 days'
    hw_category = heatwave  <- C6 'Heat wave by absolute temperature'
      threshold_met = True  <- C1 'Plains threshold reached'
        region = plains  (given by user)
        tmax = 45.6  (given by user)
      tmax = 45.6  (given by user)
    heat_days = 3  (given by user)
```

```
KNOWLEDGE SOURCES USED
--------------------------------------------------------------------
--
S1: IMD - Heat wave criteria (FAQ on Heat Wave, mausam.imd.gov.in):
threshold 40 C plains / 37 C coastal / 30 C hilly; departure from
normal 4.5-6.4 C = heat wave, >6.4 C = severe; actual Tmax >=45 C =
heat wave, >=47 C = severe; declared when met on 2 consecutive days.
S2: IMD - Colour-coded impact-based heat warnings (Green / Yellow /
Orange / Red) based on intensity and persistence of heat wave days.
S3: IMD - Warm night criteria: Tmax >=40 C and Tmin departure 4.5-6.4
C = warm night, >6.4 C = very warm night.
S4: NDMA - Guidelines for Preparation of Action Plan: Prevention and
Management of Heat Wave (2016, rev. 2019): Do's & Don'ts, stakeholder
roles, cooling centres, work-hour rescheduling.
S5: MoHFW / NCDC - National Programme on Climate Change & Human Health
(NPCCHH): National Action Plan on Heat Related Illnesses (2021) - HRI
symptoms, first aid, hospital preparedness.
S6: US NWS / NOAA - Heat Index (Rothfusz regression, 1990) and risk
bands: Caution 27-32 C, Extreme caution 32-39 C, Danger 39-51 C,
Extreme danger >=51 C.
====================================================================
==
```

D4. Test Case Report

The automated suite (`python -m pytest`) runs **23 scenarios** plus unit tests for the web API, conflict resolution, the HOW explanation, the heat-index reference values and wrong-type input. Each scenario asserts the expected conclusions, the rules that must fire and the rules that must *not* fire. Together the scenarios make **every one of the 59 rules fire at least once** (rule coverage 59/59 (100 %)). They cover all 4 alert levels, all 4 stakeholder roles, all 3 region types, all 3 heat-illness types, 6 kinds of invalid input, missing optional data, and exact threshold boundaries.

Result: 1 failed, 34 passed in 0.09s

4.1 Summary

TC	Scenario	Alert / outcome	Rules fired	Result
TC01	Normal summer day, Pune (plains) - no heat wave	GREEN / none	7	PASS
TC02	Humid heat, Mumbai (coastal) - below IMD threshold	YELLOW / none	10	FAIL
TC03	Severe heat wave for 4 days, Vidarbha (farmer)	RED / severe_heatwave	11	PASS
TC04	Heat wave criteria met on day 1 only (municipal authority)	YELLOW / heatwave	9	PASS
TC05	Hill station (Mahabaleshwar) severe departure, 2 days (health	ORANGE / severe_heatwave	8	PASS

TC	Scenario	Alert / outcome	Rules fired	Result
	dept)			
TC06	Heat stroke symptoms during heat wave (emergency)	YELLOW / heatwave + EMERGENCY	12	PASS
TC07	Invalid sensor reading (Tmax = 62 C) - graceful halt	ERROR (halt V1)	1	PASS
TC08	Missing optional data (no normal, no duration, no humidity)	YELLOW / heatwave	7	PASS
TC09	Warm night escalates Yellow to Orange (vulnerable household)	ORANGE / heatwave	15	PASS
TC10	Inconsistent data: Tmin higher than Tmax	ERROR (halt V5)	2	PASS
TC11	Boundary: Tmax exactly 40.0 C, departure exactly 4.5 C	YELLOW / heatwave	9	PASS
TC12	Boundary: 39.9 C in plains with +5 C departure - threshold not met	GREEN / none	7	PASS
TC13	Prolonged heat wave (7 days) - Red with human review	RED / heatwave	12	PASS
TC14	Coastal heat wave with very warm night (Ratnagiri)	ORANGE / heatwave	14	PASS
TC15	Heat wave day 2 with extreme humid heat index (Konkan coast)	ORANGE / heatwave	9	PASS
TC16	Classic heat stroke (hot, dry skin) reported to health dept	YELLOW / heatwave + EMERGENCY	11	PASS
TC17	Heat exhaustion on a normal day (exertional)	GREEN / none	9	PASS
TC18	Farmer on a normal day - routine advice only	GREEN / none	7	PASS
TC19	Health department on a normal day	GREEN / none	7	PASS
TC20	Invalid humidity (140 %) - graceful halt	ERROR (halt V3)	1	PASS
TC21	Implausible normal temperature (70 C) - graceful halt	ERROR (halt V2)	1	PASS
TC22	Invalid persistence (0 days) - graceful halt	ERROR (halt V4)	1	PASS
TC23	Heat wave persisting 5 days - Orange for authority	ORANGE / heatwave	9	PASS

4.2 Detailed scenarios

TC01: Normal summer day, Pune (plains) - no heat wave

- *Inputs:* `role=citizen, region=plains, tmax=34.0, tmax_normal=33.0, heat_days=1, rh=30`
- *Expected:* alert = GREEN, hw_category = none
- *Rules fired (in order):* D1 → D3 → H2 → C8 → A1 → P1 → K1
- *Output:* GREEN ALERT (No action - normal day); advice for: CITIZEN
- *Result:* **PASS**

TC02: Humid heat, Mumbai (coastal) - below IMD threshold

- *Inputs:* `role=citizen, region=coastal, tmax=35.0, tmax_normal=33.0, heat_days=2, rh=60, vulnerable=elderly`
- *Expected:* alert = YELLOW, hw_category = none, hi_level = danger, escalated = humid heat stress
- *Rules fired (in order):* D1 → D3 → C2 → H3 → C8 → A1 → E2 → P4 → P2 → K1
- *Output:* YELLOW ALERT (Heat alert - be updated); advice for: CITIZEN, VULNERABLE
- *Result:* **FAIL**

TC03: Severe heat wave for 4 days, Vidarbha (farmer)

- *Inputs:* `role=farmer, region=plains, tmax=47.2, tmax_normal=42.0, heat_days=4, rh=18, farm=both`
- *Expected:* alert = RED, hw_category = severe_heatwave, needs_review = True
- *Rules fired (in order):* D1 → D3 → C1 → H3 → C4 → A7 → G3 → G1 → F2 → F3 → K1
- *Output:* RED ALERT (Extreme heat alert - take action) - IMD category: severe heatwave; advice for: FARMER, GOVERNANCE, NOTE
- *Result:* **PASS**

TC04: Heat wave criteria met on day 1 only (municipal authority)

- *Inputs:* `role=authority, region=plains, tmax=44.0, tmax_normal=39.0, heat_days=1, rh=20`
- *Expected:* alert = YELLOW, hw_category = heatwave, status = watch
- *Rules fired (in order):* D1 → D3 → C1 → H3 → C7 → A2 → G3 → X1 → K1
- *Output:* YELLOW ALERT (Heat alert - be updated) - IMD category: heatwave [WATCH: not yet declared, needs 2nd consecutive day]; advice for: AUTHORITY, NOTE
- *Result:* **PASS**

TC05: Hill station (Mahabaleshwar) severe departure, 2 days (health dept)

- *Inputs:* `role=health, region=hilly, tmax=31.5, tmax_normal=24.5, heat_days=2, rh=40`
- *Expected:* alert = ORANGE, hw_category = severe_heatwave
- *Rules fired (in order):* D1 → D3 → C3 → H1 → C5 → A6 → M2 → K1
- *Output:* ORANGE ALERT (Severe heat alert - be prepared) - IMD category: severe heatwave; advice for: HEALTH
- *Result:* **PASS**

TC06: Heat stroke symptoms during heat wave (emergency)

- *Inputs:* `role=citizen, region=plains, tmax=45.0, tmax_normal=40.0, heat_days=2, rh=22, vulnerable=elderly, symptoms=high_body_temp/confusion/no_sweating`
- *Expected:* alert = YELLOW, hri = heat_stroke, emergency = True
- *Rules fired (in order):* T1 → T5 → D1 → D3 → C1 → H3 → C6 → A3 → G3 → P4 → P2 → K1
- *Output:* YELLOW ALERT (Heat alert - be updated) - IMD category: heatwave; advice for: CITIZEN, EMERGENCY, NOTE, VULNERABLE
- *Result:* **PASS**

TC07: Invalid sensor reading (Tmax = 62 C) - graceful halt

- *Inputs:* `role=citizen, region=plains, tmax=62.0, tmax_normal=40.0, heat_days=2`
- *Expected:* error = Tmax outside plausible range -10..55 C
- *Rules fired (in order):* V1
- *Output:* INPUT ERROR: Tmax outside plausible range -10..55 C
- *Result:* **PASS**

TC08: Missing optional data (no normal, no duration, no humidity)

- *Inputs:* `role=citizen, region=plains, tmax=46.0`
- *Expected:* alert = YELLOW, hw_category = heatwave, status = watch
- *Rules fired (in order):* C1 → C6 → A8 → G2 → G3 → P2 → K1
- *Output:* YELLOW ALERT (Heat alert - be updated) - IMD category: heatwave [WATCH: not yet declared, needs 2nd consecutive day]; advice for: CITIZEN, NOTE
- *Result:* **PASS**

TC09: Warm night escalates Yellow to Orange (vulnerable household)

- *Inputs:* `role=citizen, region=plains, tmax=45.6, tmax_normal=39.8, heat_days=3, rh=25, tmin=31.0, tmin_normal=26.0, vulnerable=elderly/outdoor_worker`
- *Expected:* alert = ORANGE, night = warm_night, escalated = warm night
- *Rules fired (in order):* D0 → D1 → D2 → D3 → C1 → W1 → H4 → C6 → A3 → E1 → P4 → P5 → P2 → P3 → K1
- *Output:* ORANGE ALERT (Severe heat alert - be prepared) - IMD category: heatwave; advice for: CITIZEN, VULNERABLE, WORKER
- *Result:* **PASS**

TC10: Inconsistent data: Tmin higher than Tmax

- *Inputs:* `role=citizen, region=plains, tmax=41.0, tmax_normal=38.0, tmin=43.0, tmin_normal=27.0`
- *Expected:* error = Minimum temperature is higher than maximum temperature
- *Rules fired (in order):* D0 → V5
- *Output:* INPUT ERROR: Minimum temperature is higher than maximum temperature
- *Result:* **PASS**

TC11: Boundary: Tmax exactly 40.0 C, departure exactly 4.5 C

- *Inputs:* `role=authority, region=plains, tmax=40.0, tmax_normal=35.5, heat_days=2, rh=15`
- *Expected:* alert = YELLOW, hw_category = heatwave
- *Rules fired (in order):* D1 → D3 → C1 → H2 → C7 → A3 → G3 → X1 → K1
- *Output:* YELLOW ALERT (Heat alert - be updated) - IMD category: heatwave; advice for: AUTHORITY, NOTE
- *Result:* **PASS**

TC12: Boundary: 39.9 C in plains with +5 C departure - threshold not met

- *Inputs:* `role=authority, region=plains, tmax=39.9, tmax_normal=34.9, heat_days=3, rh=15`
- *Expected:* alert = GREEN, hw_category = none
- *Rules fired (in order):* D1 → D3 → H2 → C8 → A1 → X3 → K1
- *Output:* GREEN ALERT (No action - normal day); advice for: AUTHORITY
- *Result:* **PASS**

TC13: Prolonged heat wave (7 days) - Red with human review

- *Inputs:* `role=health, region=plains, tmax=45.5, tmax_normal=41.5, heat_days=7, rh=20, symptoms=muscle_cramps`
- *Expected:* alert = RED, hw_category = heatwave, hri = heat_cramps, needs_review = True
- *Rules fired (in order):* T4 → T7 → D1 → D3 → C1 → H3 → C6 → A5 → G3 → G1 → M2 → K1
- *Output:* RED ALERT (Extreme heat alert - take action) - IMD category: heatwave; advice for: FIRST AID, GOVERNANCE, HEALTH, NOTE
- *Result:* **PASS**

TC14: Coastal heat wave with very warm night (Ratnagiri)

- *Inputs:* `role=citizen, region=coastal, tmax=40.0, tmax_normal=34.0, heat_days=2, rh=55, tmin=32.0, tmin_normal=25.0, vulnerable=infant_child`
- *Expected:* alert = ORANGE, hw_category = heatwave, night = very_warm_night
- *Rules fired (in order):* D0 → D1 → D2 → D3 → C2 → W2 → H4 → C7 → A3 → E1 → P4 → P2 → P3 → K1
- *Output:* ORANGE ALERT (Severe heat alert - be prepared) - IMD category: heatwave; advice for: CITIZEN, VULNERABLE
- *Result:* **PASS**

TC15: Heat wave day 2 with extreme humid heat index (Konkan coast)

- *Inputs:* `role=authority, region=coastal, tmax=38.5, tmax_normal=33.5, heat_days=2, rh=65`
- *Expected:* alert = ORANGE, hw_category = heatwave, hi_level = extreme_danger, escalated = extreme humid heat
- *Rules fired (in order):* D1 → D3 → C2 → H4 → C7 → A3 → E3 → X2 → K1

- *Output:* ORANGE ALERT (Severe heat alert - be prepared) - IMD category: heatwave; advice for: AUTHORITY
- *Result:* **PASS**

TC16: Classic heat stroke (hot, dry skin) reported to health dept

- *Inputs:* `role=health, region=plains, tmax=44.0, tmax_normal=39.0, heat_days=2, rh=20, symptoms=high_body_temp/no_sweating`
- *Expected:* alert = YELLOW, hri = heat_stroke, emergency = True
- *Rules fired (in order):* T2 → T5 → D1 → D3 → C1 → H3 → C7 → A3 → G3 → M1 → K1
- *Output:* YELLOW ALERT (Heat alert - be updated) - IMD category: heatwave; advice for: EMERGENCY, HEALTH, NOTE
- *Result:* **PASS**

TC17: Heat exhaustion on a normal day (exertional)

- *Inputs:* `role=citizen, region=plains, tmax=34.0, tmax_normal=33.0, heat_days=1, rh=30, symptoms=heavy_sweating/dizziness`
- *Expected:* alert = GREEN, hri = heat_exhaustion
- *Rules fired (in order):* T3 → T6 → D1 → D3 → H2 → C8 → A1 → P1 → K1
- *Output:* GREEN ALERT (No action - normal day); advice for: CITIZEN, FIRST AID
- *Result:* **PASS**

TC18: Farmer on a normal day - routine advice only

- *Inputs:* `role=farmer, region=plains, tmax=36.0, tmax_normal=35.0, heat_days=1, rh=35, farm=crops`
- *Expected:* alert = GREEN, hw_category = none
- *Rules fired (in order):* D1 → D3 → H2 → C8 → A1 → F1 → K1
- *Output:* GREEN ALERT (No action - normal day); advice for: FARMER
- *Result:* **PASS**

TC19: Health department on a normal day

- *Inputs:* `role=health, region=coastal, tmax=33.0, tmax_normal=32.0, heat_days=1, rh=40`
- *Expected:* alert = GREEN
- *Rules fired (in order):* D1 → D3 → H2 → C8 → A1 → M3 → K1
- *Output:* GREEN ALERT (No action - normal day); advice for: HEALTH
- *Result:* **PASS**

TC20: Invalid humidity (140 %) - graceful halt

- *Inputs:* `role=citizen, region=coastal, tmax=36.0, rh=140`
- *Expected:* error = Relative humidity must be 0..100 %
- *Rules fired (in order):* V3
- *Output:* INPUT ERROR: Relative humidity must be 0..100 %
- *Result:* **PASS**

TC21: Implausible normal temperature (70 C) - graceful halt

- *Inputs:* `role=citizen, region=plains, tmax=44.0, tmax_normal=70.0`
- *Expected:* error = Normal Tmax outside plausible range 5..50 C
- *Rules fired (in order):* V2
- *Output:* INPUT ERROR: Normal Tmax outside plausible range 5..50 C
- *Result:* **PASS**

TC22: Invalid persistence (0 days) - graceful halt

- *Inputs:* `role=authority, region=plains, tmax=44.0, heat_days=0`
- *Expected:* error = Consecutive hot days must be between 1 and 60
- *Rules fired (in order):* V4
- *Output:* INPUT ERROR: Consecutive hot days must be between 1 and 60
- *Result:* **PASS**

TC23: Heat wave persisting 5 days - Orange for authority

- *Inputs:* `role=authority, region=plains, tmax=45.5, tmax_normal=41.5, heat_days=5, rh=15`
- *Expected:* alert = ORANGE, hw_category = heatwave
- *Rules fired (in order):* D1 → D3 → C1 → H3 → C6 → A4 → G3 → X2 → K1
- *Output:* ORANGE ALERT (Severe heat alert - be prepared) - IMD category: heatwave; advice for: AUTHORITY, NOTE
- *Result:* **PASS**

4.3 Cross-check with the Prolog edition

Scenarios TC01, TC02, TC03, TC04, TC06 and TC09 were also run through the backward-chaining Prolog edition (`?- run_tests_cases.`). It produced **the same alert levels** (Green, Yellow, Red, Yellow-watch, Yellow + emergency, Orange), which shows that the knowledge is independent of the inference strategy.

D5. Demo Plan (live run, 2-3 queries)

1. **Normal vs. watch vs. declared**: TC01 (Green) → TC04 (Yellow *watch*, day 1) → change days to 3 (Yellow *declared*). Shows the IMD persistence rule.
2. **Escalation and HOW**: TC09. Show the trace, then type `how alert` to see Yellow (A3) → Orange (E1, warm night).
3. **Robustness and life safety**: enter Tmax = 62 (V1 halts with a clear message), then TC06 (heat-stroke symptoms → EMERGENCY advice fires first).
4. *(Optional)* The same consultation in Prolog: type `why` at the "normal temperature" question to show the goal stack.

D6. Reflection Notes (individual, ½ page each)

Each member writes this section separately in their own words. The prompts below are a guide.

Member 1, _____ (Roll No. _____) *Contribution:*
_____ *One limitation of rule-based systems observed:*
_____

Member 2, _____ (Roll No. _____) *Contribution:*
_____ *One limitation of rule-based systems observed:*
_____

Member 3, _____ (Roll No. _____) *Contribution:*
_____ *One limitation of rule-based systems observed:*
_____

Limitations we observed, for members to choose from and expand:

- **Brittle crisp thresholds.** 39.9 °C gives *Green* and 40.0 °C gives *Yellow* (TC11 vs. TC12). Real risk is continuous, and fuzzy logic or certainty factors would model it better.
- **The knowledge-acquisition bottleneck.** IMD defines the colour code in words ("persists for more than 2 days"), and turning these into rules forced us to make choices. Day-1 severe heat had to become a *watch*, and the case where the duration is unknown needed its own rule (A8).
- **Rule interaction is hard to reason about.** Our first version escalated hot, dry 44 °C days to Orange, because the Rothfusz heat-index formula exaggerates heat stress in very dry air. Adding the RH ≥ 40 % condition to E3 fixed it, but the bug appeared only through testing.
- **No learning.** The system cannot adapt to local mortality data, so new evidence means a manual rule edit.
- **No uncertainty handling.** A forecast of 45 °C with 60 % confidence is treated the same as an observed 45 °C.
- **Scalability.** Validating 59 rules was manageable, but with hundreds of rules, conflicts and redundancy become hard to spot without tools.

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